

GeoTrace Forensics

Investigation Guide

A Step-by-Step Forensic Workflow Manual

Digital Image Metadata & Geolocation Analysis

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1. Introduction

This document is a practical guide for digital forensic investigators using GeoTrace Forensics, a desktop application designed to extract, analyze, and visualize hidden metadata in digital images.

Modern smartphones and cameras automatically embed information into every photograph they take. This metadata includes the camera model, exact date and time, GPS coordinates, software used, and editing history. While invisible to the average viewer, this information is invaluable for investigators trying to verify alibis, place suspects at crime scenes, or detect fabricated evidence.

GeoTrace Forensics consolidates the entire forensic image analysis workflow into a single tool, from evidence intake through to court-ready reporting.

Who This Guide Is For

- Digital forensic investigators
- Law enforcement personnel handling digital evidence
- Cybersecurity professionals
- Students learning practical digital forensics

What You Will Learn

- How to create and manage forensic cases
- How to import and analyze evidence images
- How to interpret EXIF metadata, GPS data, and timestamps
- How to detect image manipulation using AI techniques
- How to maintain a tamper-proof chain of custody
- How to generate professional forensic reports

2. Tool Overview

GeoTrace Forensics is built around a clear, tab-based interface. Each tab represents a different aspect of the investigation.

Main Components

Component	Purpose
Case Management Panel	Create new cases or open existing ones
Investigation Actions Bar	Primary actions: add images, generate maps, create reports, verify integrity
Results Table	Lists all evidence images in the current case with quick status info
Detail Tabs	Drill-down into individual image details across six specialized views

The Six Detail Tabs

Tab	Purpose	When to Use
Preview	Visual image preview	Identify images quickly
Metadata	Key forensic fields	Quick summary of EXIF data
Anomalies	Suspicious findings	Spot tampering signs
Raw EXIF	Complete EXIF dump	Deep technical analysis
Timeline	Chronological events	Build movement story
AI Analysis	Manipulation detection	Verify image authenticity

There is also a Blockchain tab dedicated to verifying the integrity of the evidence log itself.

3. Forensic Workflow at a Glance

Every investigation in GeoTrace Forensics follows the same fundamental workflow:

1. Create a new case with a unique name and the investigator's full name.
2. Import evidence images. The tool automatically calculates SHA-256 hashes, extracts EXIF, detects anomalies, and logs each file to the blockchain.
3. Review the results table for an overview of all evidence.
4. Click on any image to inspect its details across the six tabs.
5. Generate the interactive map to visualize geolocation data and movement patterns.
6. Run AI manipulation analysis on suspicious images.
7. Verify the blockchain to confirm chain-of-custody integrity.
8. Generate a PDF forensic report for case files or court submission.

Note: Every action in GeoTrace is non-destructive. The original evidence files are never modified. All analysis is performed on copies and recorded hashes.

4. Sample Case Scenario: The Suspicious Alibi

Background

On the evening of August 13, 2009, a burglary occurred at a private residence in Florence, Italy. A suspect, Mr. Rossi, was arrested. During interrogation, he claimed he was at home in another city the entire day and provided five photographs from his phone as proof.

Detective Selim, a digital forensic investigator, has been assigned to verify the authenticity of these images. His goal is to determine:

- Were these photos actually taken on August 13, 2009?
- Were they taken at the location Mr. Rossi claims?
- Have any of the photos been edited or tampered with?
- Can the photos be linked to other devices or sources?

The Evidence

Detective Selim receives the following images for analysis:

File Name	Stated by Suspect	Investigator Notes
DSCN0010.jpg	Taken at home	Camera: Nikon Coolpix
DSCN0012.jpg	Taken at home	Same camera
DSCN0029.jpg	Taken at home	Same camera
tampered_1.jpg	Personal photo	Suspicious quality
tampered_2.jpg	Personal photo	Suspicious quality

Detective Selim now uses GeoTrace Forensics to analyze each image.

5. Step-by-Step Investigation

Step 1: Launch the Application

Open a terminal in the project folder and run:

```
python main.py
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The GeoTrace Forensics window opens, displaying the El Sewedy University logo and the Case Management panel.

Step 2: Create a New Case

- 9. In the Case Management panel, type the case name: Rossi Burglary Investigation
- 10. Type the investigator name: Detective Selim ElSaadany
- 11. Click Create Case.

A confirmation appears with a unique case ID. The case is now active and ready for evidence intake.

Tip: Always use a clear, descriptive case name that includes the suspect or location. This helps when reopening cases later.

Step 3: Import Evidence Images

- 12. Click the Add Images button.
- 13. In the file dialog, select all five JPG files from the suspect's phone.
- 14. Click Open.

GeoTrace performs the following actions automatically for each image:

- Calculates a SHA-256 hash to lock in file integrity.
- Extracts complete EXIF metadata.
- Decodes GPS coordinates if available.
- Runs anomaly detection.
- Adds a new block to the evidence blockchain.
- Stores everything in the case database.

A confirmation message displays the number of images successfully imported.

Step 4: Review the Results Table

The results table now lists all five images. Each row shows essential forensic fields at a glance:

Column	What It Shows	Why It Matters
Image	File name	Identification
Capture Time	EXIF date taken	Verifies alibi timing
Coordinates	Decoded GPS	Verifies alibi location

Camera	Make and model	Device fingerprint
Integrity	Hash match status	File untouched?
Anomalies	Issue count	Suspicion level

Step 5: Generate the Interactive Map

Click the Generate Map button. A new browser tab opens displaying:

- Pin markers for every image with valid GPS coordinates.
- A movement path connecting locations chronologically.
- A timeline slider that animates the photos in time order.
- A case summary box with totals.
- A legend explaining marker colors.

Detective Selim observes that three of the five images contain GPS data, all clustered in Florence, Italy — not at Mr. Rossi's home as claimed. The other two (the tampered_*.jpg files) have no GPS data at all.

Note: The absence of GPS in the tampered images is itself a red flag. Modern smartphones almost always record location unless the file has been re-saved or processed through editing software.

6. Interpreting Results

Reading EXIF Metadata

Click any image in the results table, then click the Metadata tab. The most forensically important fields are:

Field	Forensic Significance
Camera Make / Model	Identifies the device. Mismatches across photos may indicate fabrication.
Date Taken	When the camera recorded the image. Compare against suspect statements.
GPS Coordinates	Where the photo was taken. Verifiable via Google Maps.
Software	Editing software leaves traces. Photoshop, GIMP, Lightroom are red flags.
Image Dimensions	Should match camera specs. Unusual sizes suggest cropping or re-export.
Date Modified	If different from Date Taken, the file was re-saved after capture.

Understanding Anomalies

The Anomalies tab lists every suspicious finding for the selected image. Common anomalies include:

- Missing camera make or model — typical of edited or re-saved files.
- Editing software detected — direct evidence of post-capture modification.
- Missing GPS data — phones almost always record this; absence may mean stripping.
- Date Taken differs from Date Modified — sign the file was re-saved.
- Missing thumbnail — editing programs often remove embedded thumbnails.
- Duplicate Evidence — the same hash appears in multiple files, indicating identical or copied images.

Anomaly Severity Levels

Count	Interpretation	Recommended Action
0	Image likely authentic	Accept as evidence
1-2	Minor concerns	Document and continue investigation

3+	Strong manipulation signals	Run AI Analysis, treat with caution
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7. AI Manipulation Analysis

For images with multiple anomalies or simply when authenticity is in question, GeoTrace provides an AI-powered manipulation detector. This goes beyond metadata to examine the actual pixel content of the image.

How to Run AI Analysis

- 15. Click on the suspect image in the results table.
- 16. Switch to the AI Analysis tab.
- 17. Click Run AI Analysis.

The tool runs three forensic techniques in sequence and produces a verdict.

The Three Detection Techniques

1. Error Level Analysis (ELA)

ELA re-saves the image at a known compression quality and compares it to the original. Edited regions compress differently from untouched regions, appearing as bright spots in the ELA visualization.

- A clean photo shows uniform, low-intensity noise.
- An edited photo shows bright clusters around the modified areas.

2. Noise Consistency Analysis

Authentic camera images contain consistent sensor noise across the entire frame. Edited or AI-generated images often have inconsistent noise patterns. The tool divides the image into four quadrants and compares their noise levels.

3. Metadata Cross-Validation

Cross-checks EXIF fields for internal consistency:

- Editing software tags (Photoshop, GIMP, etc.)
- Date Taken vs Date Modified mismatches
- Phone camera with stripped GPS
- Missing thumbnails

Interpreting the AI Verdict

Verdict	Confidence Range	Recommended Action
Likely Authentic	0-25%	Accept as evidence
Possibly Modified	25-50%	Investigate further
Probably Manipulated	50-75%	Treat as suspicious
Almost Certainly Manipulated	75-100%	Reject or seek original

Note: The AI analysis is a screening tool, not a definitive judgment. Always combine its output with metadata findings and contextual investigation.

8. Blockchain Chain of Custody

Chain of custody is the documented record of who handled evidence and when. In digital forensics, this is critical for court admissibility. GeoTrace implements chain of custody using a lightweight blockchain.

How the Blockchain Works

Every time an image is added to a case, a new block is created containing:

- The file name.
- The file's SHA-256 hash.
- The investigator name.
- The exact timestamp.
- A cryptographic link to the previous block.

Each block's ID is computed by hashing all of its fields together. If anyone modifies any field, the block ID will no longer match — and because each block links to the previous block's ID, the entire chain after that point breaks visibly.

Verifying the Chain

18. Click the Blockchain tab.
19. Click Verify Chain.
20. Read the verdict at the top of the panel.

Verdict	Meaning
Chain Intact	All evidence records match their cryptographic signatures. Evidence is legally pristine.
Chain BROKEN	At least one block has been modified. The exact failed block is identified. Evidence integrity is compromised.
No Evidence Logged Yet	The case is empty. Add images first.

What Tampering Detection Looks Like

If a malicious actor modifies the evidence log file directly (for example, to change the recorded hash of a file or remove an entry), the next chain verification will:

- Mark the affected block as TAMPERED.
- Identify all subsequent blocks as having BROKEN LINK status.
- Display the detected mismatch in the verification log.

Tip: During presentations or court demonstrations, the live tampering test is highly effective: show Chain Intact, manually edit the blockchain.json file, then re-verify to demonstrate immediate detection.

9. Generating the Final Report

Once analysis is complete, GeoTrace can produce a professional PDF report suitable for case files or court submission.

How to Generate the Report

21. Ensure the case is loaded and all evidence has been analyzed.
22. Click Generate PDF Report from the action bar.
23. The PDF opens automatically in your default viewer.

What the Report Contains

- Cover page with case name, investigator, and date.
- Case summary with totals: images, GPS-tagged, anomalies, distance traveled.
- Evidence chain table with file names, paths, hashes, integrity status, and duplicates.
- Timeline findings: every image with capture time, coordinates, and analysis.
- Correlated images: identifying photos taken close together in time or location.
- Anomaly register: every suspicious finding documented with severity.

Additional Export Formats

Beyond the PDF, two additional exports are available:

- Export Excel — full case data in spreadsheet form for further analysis.
- Export JSON — machine-readable format for integration with other tools.

10. Best Practices for Investigators

Evidence Handling

- Always work on copies of evidence files. Never modify originals.
- Calculate SHA-256 hashes immediately upon receiving files. GeoTrace does this automatically.
- Document every step in case notes. The PDF report supports this but does not replace it.
- Keep the blockchain file backed up separately from the case files.

Case Management

- Use descriptive case names: 'Rossi Burglary 2009' is better than 'Test 1'.
- Store the investigator's full legal name, not nicknames.
- Re-verify the blockchain at the start of each session.

Image Analysis

- Always review anomalies before drawing conclusions.
- Run AI Analysis on any image with 2 or more anomalies.
- Cross-reference GPS coordinates with mapping services to confirm locations.
- Note time zones — EXIF timestamps may be in local time, not UTC.

Reporting

- Generate the PDF report at the end of each significant analysis session.
- Include both authentic and manipulated images in the report for comparison.
- Archive the report alongside the original evidence files and the blockchain log.

11. Troubleshooting

Problem	Solution
No GPS data found	The image's EXIF may have been stripped. This itself is an anomaly. Document it.
Map shows no pins	None of the imported images have GPS coordinates. Add GPS-enabled images.
AI Analysis fails	Check the file is a supported format (JPG works best). PNG and other formats may have limited ELA accuracy.
Chain Broken on first run	If you see this on a fresh case, restart the app and re-import. May indicate a write error.
PDF report won't open	Ensure a default PDF viewer is installed. Try opening the file directly from the output folder.
Application won't start	Verify Python 3.10+ is installed. Run <code>pip install -r requirements.txt</code> to install all dependencies.

12. Glossary

Term	Definition
EXIF	Exchangeable Image File Format. Metadata embedded in image files by cameras and editing software.
SHA-256	A cryptographic hash function. Produces a unique 64-character fingerprint for any file.
GPS Rationals	The raw EXIF format for storing GPS coordinates as fractions. Must be decoded into decimal degrees.
ELA	Error Level Analysis. A technique that reveals image manipulation by comparing JPEG compression levels.
Chain of Custody	The legal record of who handled evidence, when, and what was done with it.
Genesis Block	The first block in a blockchain. Has no predecessor.
Anomaly	Any deviation from expected metadata patterns that may indicate tampering.
Manipulation	Any post-capture editing of an image, whether benign (cropping) or malicious (forgery).
Forensic Soundness	The principle that evidence handling must not alter the evidence in any way.